[CSL252]-Design and Analysis of Algorithms **2025-26-W**

TUTORIAL IX

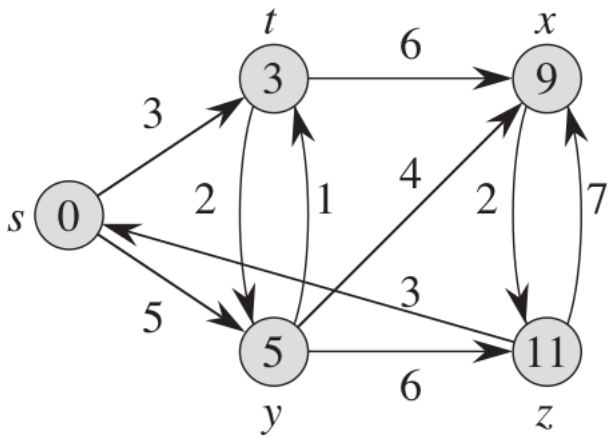
Date of the tutorial: **Mar 19, 2026**.

Due Date: **11:59 pm, Mar 23, 2026**.

Instructions. Same as Tutorial I.

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1. Run the Dijkstra's algorithm on the following weighted directed graph and find the shortest distance from source s to every other vertex. Draw the shortest-path tree.



2. Run the Dijkstra's algorithm on the above graph using priority queue implementation. Show the priority queue after every step.

3. Give an example of a directed graph with negative-weight edges for which Dijkstra's algorithm produces incorrect answers.

4. (a) Consider a directed graph which has exactly one negative edge leaving s; all other edges are positive. Can Dijkstra's algorithm, started at s, fail on such a graph? justify your answer.

   (b) Consider a directed graph in which the only negative edges are those that leave s; all other edges are positive. Can Dijkstra's algorithm, starting at s, fail on such a graph? justify your answer.

5. Let G be a weighted (edge weights may be negative) directed acyclic graph with a source vertex s.

   (a) Design an algorithm to compute the length of the shortest path from s to every other vertex of G.

   (b) Design an algorithm to compute the length of the longest path from s to every other vertex of G.

6. (a) How can we use Floyd-Warshall algorithm to determine whether there exists a negative weight cycle in the graph. Clearly write your proof.

   (b) How can we use Bellman-Ford algorithm to determine whether there exists a negative weight cycle in the graph. Clearly write your proof.

7. Run the Floyd-Warshall algorithm on the following edge weighted, directed graph. Show the (shortest distance) matrix $D^{(k)}$ and predecessor matrix $\Pi^{(k)}$ that results after each iteration of the outer loop. Also, explain how to construct shortest paths using matrix $\Pi^{(k)}$.

